File Permission and Ownership in Linux

Every file and directory has:

- **Owner (User)**
- **Group**
- **Others You can check using:**

ls -l

```
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Feb 26 13:02:27 2026 from 3.16.146.5
[ec2-user@ip-172-31-8-144 ~]$ sudo su
[root@ip-172-31-8-144 ec2-user]# exit
exit
[ec2-user@ip-172-31-8-144 ~]$ mkdir folder1
[ec2-user@ip-172-31-8-144 ~]$ ls-l
-bash: ls-l: command not found
[ec2-user@ip-172-31-8-144 ~]$ ls
folder  folder1
[ec2-user@ip-172-31-8-144 ~]$ ls -l
total 0
drwxr-xr-x. 3 root     root     34 Feb 26 13:15 folder
drwxr-xr-x. 2 ec2-user ec2-user  6 Mar  3 01:15 folder1
[ec2-user@ip-172-31-8-144 ~]$ cd folder
[ec2-user@ip-172-31-8-144 folder]$ ls
file2  folder2
[ec2-user@ip-172-31-8-144 folder]$ ls -l
total 0
-rw-r--r--. 1 root root  0 Feb 26 13:03 file2
drwxr-xr-x. 2 root root 32 Feb 26 13:18 folder2
[ec2-user@ip-172-31-8-144 folder]$ rmd folder2
```

Breakdown:

- **ec2-user → Owner**
- **ec2-user → Group**
- **Others → Everyone else**

File permission in linux

- Permissions control what actions can be performed.
- There are **3 types of permissions**:

Symbol Number Meaning r
4 Read w
2 Write x 1
Execute **Add them:**

- **7 → 4+2+1 (rwx)**
- **6 → 4+2 (rw-)**
- **5 → 4+1 (r-x)**

- **4 → 4 (r--)**

```
[root@ip-172-31-8-144 folder2]# chmod g+w file2 file3
[root@ip-172-31-8-144 folder2]# ls -l
total 8
-rw-rw-r--. 1 root root 3 Feb 26 13:18 file2
-rw-rw-r--. 1 root root 3 Feb 26 13:16 file3
[root@ip-172-31-8-144 folder2]# chmod 777 file2 file3
[root@ip-172-31-8-144 folder2]# ls -l
total 8
-rwxrwxrwx. 1 root root 3 Feb 26 13:18 file2
-rwxrwxrwx. 1 root root 3 Feb 26 13:16 file3
[root@ip-172-31-8-144 folder2]# exit
exit
```

- **Change ec2-user to root using the command (Chown root:root filename)**

```
[ec2-user@ip-172-31-8-144 folder]$ chmod 124 file2
chmod: changing permissions of 'file2': Operation not permitted
[ec2-user@ip-172-31-8-144 folder]$ sudo su
[root@ip-172-31-8-144 folder]# chmod 124 file2
[root@ip-172-31-8-144 folder]# ls -l
total 0
---x-w-r--. 1 root root  0 Feb 26 13:03 file2
drwxr-xr-x. 2 root root 32 Feb 26 13:18 folder2
[root@ip-172-31-8-144 folder]# chown ec2-user:ec2-user file2
[root@ip-172-31-8-144 folder]# ls-lrt
bash: ls-lrt: command not found
[root@ip-172-31-8-144 folder]# ls -l
total 0
---x-w-r--. 1 ec2-user ec2-user  0 Feb 26 13:03 file2
drwxr-xr-x. 2 root     root     32 Feb 26 13:18 folder2
[root@ip-172-31-8-144 folder]# chown ec2-user:ec2-user folder2
[root@ip-172-31-8-144 folder]# ls -l
total 0
---x-w-r--. 1 ec2-user ec2-user  0 Feb 26 13:03 file2
drwxr-xr-x. 2 ec2-user ec2-user 32 Feb 26 13:18 folder2
[root@ip-172-31-8-144 folder]# exit
```

- **Add the user and Group**

```
[root@ip-172-31-8-144 ec2-user]# adduser justin
[root@ip-172-31-8-144 ec2-user]# pwd
/home/ec2-user
[root@ip-172-31-8-144 ec2-user]# cd /home
[root@ip-172-31-8-144 home]# ls
ec2-user  justin
[root@ip-172-31-8-144 home]# cat /etc/passwd | justin
bash: justin: command not found
[root@ip-172-31-8-144 home]# cat /etc/passwd | grep justin
justin:x:1001:1001::/home/justin:/bin/bash
[root@ip-172-31-8-144 home]# groupadd justin1
[root@ip-172-31-8-144 home]# cat /etc/group | grep justin1
justin1:x:1002:
```

- **Password setting to the user**

```
[root@ip-172-31-8-144 home]# passwd justin
Changing password for user justin.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-8-144 home]# cat etc/shadow | grep justin
cat: etc/shadow: No such file or directory
[root@ip-172-31-8-144 home]# cat /etc/shadow | grep justin
justin:$6$kS.sUJAw4N4z86ai$gJMF1Tbmj/48yBY8UqB5TrbNrUN8PsV104iffEPX4T1gWn5NcAZtK8VOUd0UXpo/bdol7NvIOlyvpjv0Bl8230:20515:0:99999:7:::
```